

EcoAccess Command

AI-Powered Decision Intelligence for Zero-Pollution Events

Project Vision

EcoAccess Command helps organizers conduct large public events with the lowest possible environmental impact using Google Cloud AI to predict energy demand, monitor waste, and recommend actions that reduce pollution.

Problem

Large events suffer from high energy consumption, carbon emissions, poor waste segregation, and slow operational decision-making because data is spread across disconnected systems.

Minimal Prototype Features

- **Carbon & Energy Prediction (BigQuery ML):** Predict energy demand and recommend actions.
- **Waste Detection (Vertex AI Vision):** Detect overflowing bins and recycling contamination.
- **AI Decision Copilot (Gemini):** Explain issues and recommend sustainability actions.
- **Knowledge Assistant (AlloyDB + pgvector):** Answer questions using uploaded sustainability guidelines.
- **Deployment (Cloud Run):** Host the application.

Demo Workflow

1. Simulated event data arrives.
2. BigQuery ML predicts higher energy usage.
3. Vision AI detects waste issues.
4. Gemini analyzes the situation.
5. RAG retrieves guidelines.
6. Dashboard recommends actions to reduce pollution.

Expected Impact

Lower carbon emissions, improved recycling efficiency, faster operational decisions, and a reusable SaaS platform for sustainable events.

Tagline

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